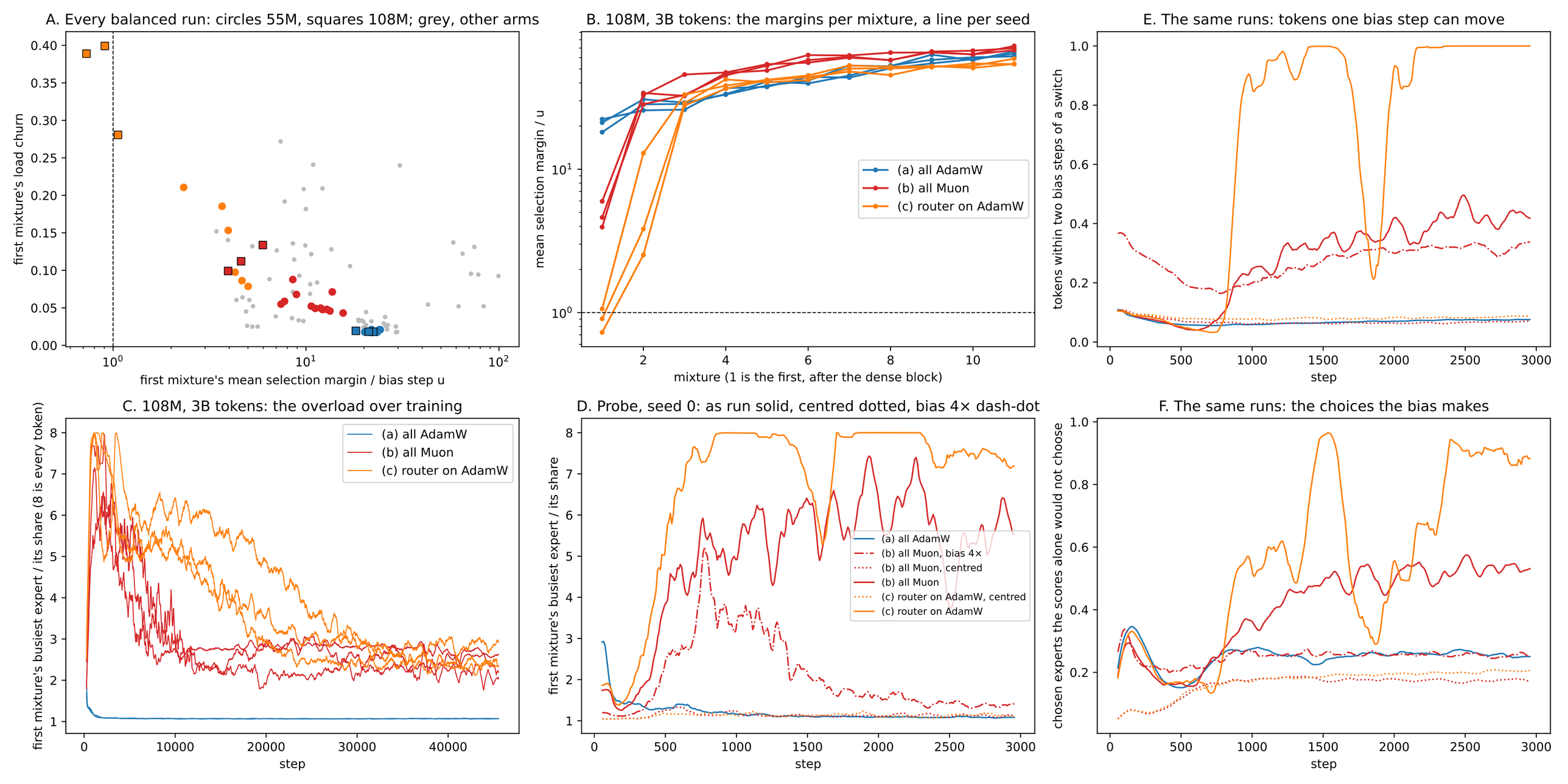

A. Every balanced run: circles 55M, squares 108M; grey, other arms
first mixture's load churn
first mixture's mean selection margin / bias step u
B. 108M, 3B tokens: the margins per mixture, a line per seed
mean selection margin / u
mixture (1 is the first, after the dense block)
(a) all AdamW
(b) all Muon
(c) router on AdamW
E. The same runs: tokens one bias step can move
tokens within two bias steps of a switch
step
C. 108M, 3B tokens: the overload over training
first mixture's busiest expert / its share (8 is every token)
step
D. Probe, seed 0: as run solid, centred dotted, bias 4× dash-dot
first mixture's busiest expert / its share
(a) all AdamW
(b) all Muon, bias 4×
(b) all Muon, centred
(b) all Muon
(c) router on AdamW, centred
(c) router on AdamW
step
F. The same runs: the choices the bias makes
chosen experts the scores alone would not choose
step